

Text and Graph Memory on PersonaMem-v2 and LoCoMo

A paired, reproducible pilot comparing current Mem0 natural-language memory with the historical open-source Mem0⁹ graph pipeline under identical samples, retrieval depth, model, and answer-generation protocol.

Seed 42

Top-k 5

DeepSeek V4 Flash · non-thinking

BGE-small · 384 dimensions

0 final case failures

Executive finding. Both memory representations improve PersonaMem-v2 MCQ accuracy over the paired no-memory control, but neither handles forget requests reliably. On LoCoMo, text memory substantially outperforms graph memory: the graph returns five triples for every query yet those triples rarely contain answer evidence, producing frequent abstention. The strongest actionable finding is therefore not that graph memory is universally ineffective, but that the historical Mem0⁹ extraction and retrieval pipeline is poorly matched to these implicit-preference and dated conversational QA settings without further graph construction or retrieval improvements.

At a glance

Final corrected runs only; earlier smoke and aborted runs are excluded.

Final evaluated items

164

30 + 30 + 52 + 52

Final case failures

0

Both datasets, both backends

PersonaMem text lift

+23.3

percentage points vs control

LoCoMo text-graph gap

+14.7

F1 points; paired CI excludes 0

Study design

Paired inputs and answer conditions isolate the memory representation as far as this software-version comparison permits.

ELEMENT	PERSONAMEM-V2	LOCOMO
Sample	30 balanced text rows: five per eligible preference type	3 conversations, first 10 dated sessions, 52 evidence-eligible questions
Memory input	Only <code>related_conversation_snippet</code> ; gold preference and answers forbidden	Same complete dated sessions for both backends
Primary metric	Exact MCQ accuracy and paired lift over empty-memory control	Official LoCoMo lexical F1 and paired lift over empty-memory control
Categories	Six non-sensitive, non-multimodal preference types	1 single-hop, 2 temporal, 3 multi-hop, 4 open-domain; category 5 excluded
Uncertainty	Wilson interval for accuracy; paired bootstrap for lift	Question-level percentile bootstrap for F1 and paired differences
Shared controls	Seed 42, top-k 5, DeepSeek V4 Flash non-thinking, temperature 0, identical option/question order	

PersonaMem-v2 results

Oracle-related-snippet pilot; 30 paired rows, not the full 32k/128k long-context benchmark.

Text MCQ accuracy

43.3%

95% CI 27.4–60.8

13/30 correct; no-memory control 6/30.

Graph MCQ accuracy

40.0%

95% CI 24.6–57.7

12/30 correct; no-memory control 6/30.

Text minus graph

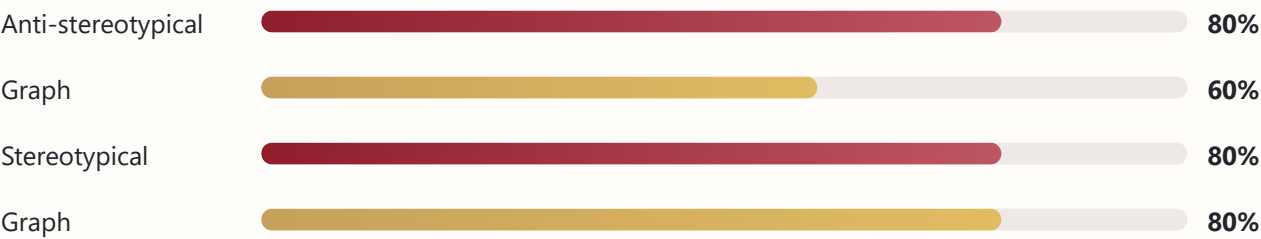
+3.3

paired 95% CI –10.0 to +16.7

No reliable overall winner on this sample.

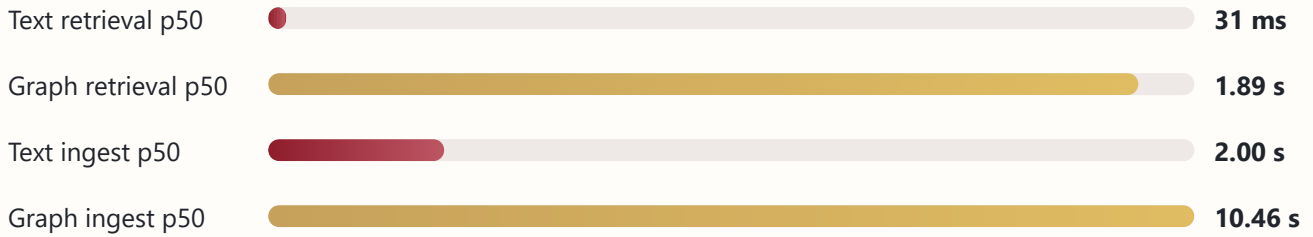
METRIC	TEXT · V2.0.14	GRAPH · V0.1.45	INTERPRETATION
Accuracy lift over no-memory	+23.3 pp [6.7, 40.0]	+20.0 pp [3.3, 36.7]	Both intervals exclude zero on this balanced pilot.
Non-empty retrieval coverage	80.0%	66.7%	Graph fails to return usable query-linked triples more often.
Non-empty storage coverage	80.0%	76.7%	Some dialogues yield no extracted memory.
Mean gold-preference token recall	19.1%	12.1%	Diagnostic only; implicit evidence need not repeat gold wording.
Forget-request MCQ accuracy	0 / 5	0 / 5	Neither backend correctly handles the natural deletion cases end-to-end.
Exact previous-preference phrase leakage	0 / 5	0 / 5	Absence of the exact phrase does not establish semantic forgetting.
Preferences about others	1 / 5	0 / 5	Both systems struggle when the preference concerns another person.

Accuracy by preference type



Other 5-row categories: health 20%/40%, neutral 40%/20%, therapy 40%/40%, forget 0%/0% (text/graph).

Latency



p95 retrieval: 45.7 ms text, 2.14 s graph. p95 ingestion: 3.49 s text, 27.15 s graph.

LoCoMo results

52 questions paired exactly across three conversations and ten dated sessions per conversation.

Text mean official F1

16.9%

95% CI 11.9–22.6

Graph mean official F1

2.2%

95% CI 0.2–4.7

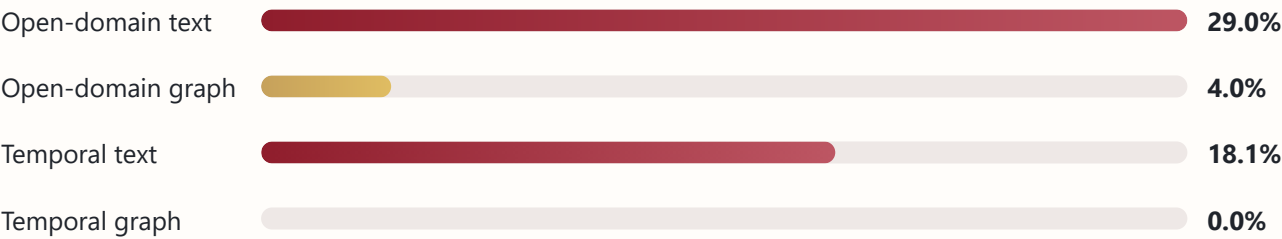
Paired text–graph gap

+14.7

95% CI +9.3 to +20.5

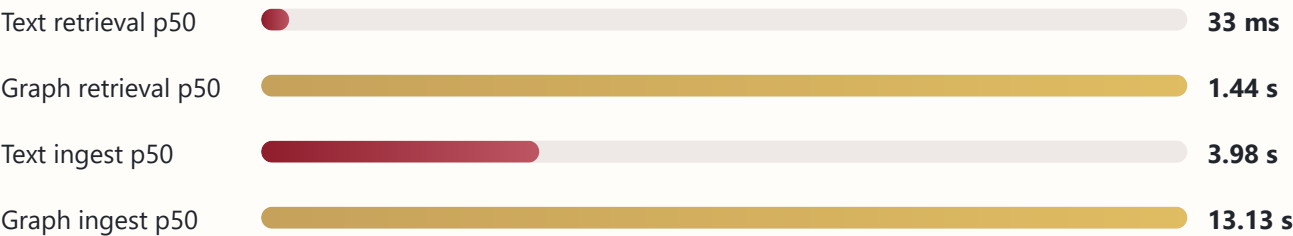
METRIC	TEXT · V2.0.14	GRAPH · V0.1.45	INTERPRETATION
Mean F1 lift over no-memory	+16.6 points [11.6, 22.3]	+1.9 points [0.0, 4.4]	Text clearly benefits from memory; graph lift is small and its CI touches zero.
Mean answer-token recall in retrieval	62.0%	3.3%	The historical graph retrieves triples, but rarely the answer-bearing content.
Answer abstention	22 / 52	49 / 52	Graph context was usually insufficient for a grounded answer.
Retrieval result count	5 / query	5 / query	Graph weakness is relevance, not empty result lists.
Question failures	0 / 52	0 / 52	The corrected evaluation harness completed every paired question.

Official F1 by category



Single-hop: 11.4% text vs 3.8% graph. Multi-hop: 1.4% text vs 0.0% graph.

Latency



Answer generation p50: 0.97 s text, 1.06 s graph. Retrieval p95: 53 ms text, 1.64 s graph.

Reliability and implementation audit

Corrections made before selecting the final result files.

1 Graph compatibility

Current `mem0ai 2.0.14` no longer exposes `graph_store`. Graph evaluation therefore runs in an isolated environment pinned to the official historical `MemoryGraph` implementation in v0.1.45.

2 DeepSeek tool JSON

DeepSeek V4 thinking mode caused malformed tool arguments during early aborted runs. Final runs explicitly use the official non-thinking toggle for both memory extraction and answer generation.

3 Safe graph boundary

Generated Neo4j labels and relationship types are normalized into valid identifiers. Malformed extraction becomes empty memory rather than invalid Cypher or a hidden retry.

Audit trail

- 22 automated tests pass.
- All final text/graph samples match exactly by row or question ID.
- Every final case completed; unparseable answer count is zero.
- All final-run Neo4j user scopes were deleted and verified empty.
- Raw reports are preserved with SHA-256 hashes in the manifest.

Operational observations retained

- Empty graph query extraction is counted as reduced coverage.
- One malformed graph deletion-candidate structure was safely treated as no-op; the run continued and cleanup succeeded.
- No result was rerun selectively to improve its score.
- Earlier smoke, failed, and aborted reports remain outside the final manifest.

What these results support—and what they do not

Supported statements

- On this paired PersonaMem pilot, both memory backends improve MCQ accuracy over the same empty-memory control.
- Neither backend successfully answers the five natural forget-request cases.
- On this paired LoCoMo pilot, text memory has substantially higher official F1 and answer-evidence recall than historical graph memory.
- The graph baseline is materially slower at ingestion and retrieval.
- Historical graph retrieval relevance is the main observed LoCoMo failure mode.

Not supported yet

- This is not a full reproduction of the Mem0 paper or either complete benchmark.
- PersonaMem uses the oracle related snippet, not the linked 32k/128k histories.
- The balanced PersonaMem sample is not prevalence-weighted.
- LoCoMo covers 3/10 conversations and ten sessions, with session-batched ingestion rather than message-pair ingestion.
- Question-level bootstrap intervals do not model clustering by persona or conversation.
- The text-graph PersonaMem difference is inconclusive; its paired CI crosses zero.

Reproduce and inspect

Run from the repository root.

```
# Validate code and services
uv run pytest -q
uv run --project legacy_graph python legacy_graph/run_personamem_graph.py --check-only

# PersonaMem-v2: identical balanced 30-row sample
uv run python scripts/run_personamem_text.py --limit 30 --seed 42
uv run --project legacy_graph \
  python legacy_graph/run_personamem_graph.py --limit 30 --seed 42

# LoCoMo: identical 3-conversation, 10-session, 52-question sample
uv run python scripts/run_locomo_text.py \
  --conversations 3 --sessions 10 --questions-per-category 5 --seed 42
uv run --project legacy_graph python legacy_graph/run_locomo_graph_v2.py \
  --conversations 3 --sessions 10 --questions-per-category 5 --seed 42
```

Final raw artifacts: [PersonaMem text](#) · [PersonaMem graph](#) · [LoCoMo text](#) · [LoCoMo graph](#) · [integrity manifest](#).

Recommended next experiment

Focus the research contribution on graph construction and retrieval alignment. The next controlled ablation should compare: (a) historical entity/relation extraction, (b) preference-aware graph schemas with explicit subject and temporal/delete edges, and (c) hybrid retrieval that combines text facts with graph neighborhoods. Keep the same paired no-memory control and measure whether the graph improvement appears specifically on multi-hop, temporal, update, and **others** cases.

Primary sources. [Mem0 paper](#); [PersonaMem-v2 paper](#); [PersonaMem-v2 dataset](#); [official LoCoMo code and evaluator](#); [DeepSeek thinking-mode documentation](#).

Confidence intervals are 95%. Percentage-point intervals for paired lifts use 5,000 deterministic bootstrap resamples with seed 42. Accuracy intervals use Wilson scores. This report intentionally avoids claiming equivalence with published Mem0 scores because the model, software versions, sample scope, and ingestion batching differ.

Prepared for research discussion with Professor Wu · Paired Mem0 baseline evaluation · Final artifacts selected 26 July 2026